

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 1 : Session – 6 : SLO – 1 | Matrix Multiplication with DMA — Allocation

■ I. Fill in the Missing Code

1. Declarations: `int **mat1, **mat2, **res; int i, j, k; int r1, c1, r2, c2;`
2. Looping variables: `int i, j, k;`
3. Dimensions: `printf("Enter rows & cols of mat1 and mat2"); scanf("%d %d %d %d", &r1, &c1, &r2, &c2);`
4. Validity check: `if (c1 != r2) { printf("Multiplication not possible\n"); return 0; }`
5. mat1: `mat1 = (int**) malloc(r1 * sizeof(int*)); for (i = 0; i < r1; i++) mat1[i] = (int*) malloc(c1 * sizeof(int));`
6. mat2: `mat2 = (int**) malloc(r2 * sizeof(int*)); for (i = 0; i < r2; i++) mat2[i] = (int*) malloc(c2 * sizeof(int));`
7. res: `res = (int**) malloc(r1 * sizeof(int*)); for (i = 0; i < r1; i++) res[i] = (int*) calloc(c2, sizeof(int));`

■ II. Notes

- malloc for input matrices (values will be read in), calloc for the result so accumulators start at 0.
- Always check `c1 == r2` before allocating the result matrix.